

Xianliang Li

PhD student at the Institute of Statistical Mathematics

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Tokyo, Japan

Research Interests

Optimizer, Foundation Model Training, Deep Learning Theory.

Education

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|---------------------------|---|
| 2025 – 2028
(Expected) | The Graduate University for Advanced Studies, SOKENDAI
Ph.D. in Statistical Science (Supervisor: Prof. Han Bao) |
| 2022 – 2025 | University of Chinese Academy of Sciences
M.Eng. in Computer Technology (Supervisor: Prof. Sheng Xu) |
| 2018 – 2022 | Sun Yat-sen University
B.Sc. in Physics |

Research Experience

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| October 2025 –
Present | The Institute of Statistical Mathematics
Research Assistant (Host: Prof. Han Bao) |
| July 2025 –
Present | The Chinese University of Hong Kong
Visiting Student (Host: Prof. Weiyang Liu) |
| September 2023
– June 2025 | Shenzhen Institutes of Advanced Technology
Research Student (Host: Prof. Sheng Xu) |

Honors and Scholarships

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| 2025 – Present | SOKENDAI Special Researcher (The Graduate University for Advanced Studies, SOKENDAI)
Supported by JST SPRING, Japan Grant Number JPMJSP2104. |
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Publications

- 2026 **Denoise First, Orthogonalize Later: Understanding Momentum in Muon via Spectral Filtering**
Xianliang Li, Zihan Zhang, Weiyang Liu, Han Bao.
Technical Report.
- 2026 **Optimal Sensor Placement Using Decentralized TDOA Sensor Network in Source Localization**
Kang Tang, Sheng Xu, Xianliang Li, K. C. Ho, He Kong, Kutluyil Dogancay.
IEEE Transactions on Aerospace and Electronic Systems.
- 2025 **3D AOA Target Tracking Using Multiple Distributed UAVs with Unknown Self-Localizations and Dynamic Anchors**
Rongrong Xu, Sheng Xu, Xianyong Mu, Xianliang Li, Dashuai Wang, Tiantian Xu.
IEEE International Conference on Real-time Computing and Robotics.
- 2025 **Time-of-Arrival Simultaneous Sensor and Target Localization with Dynamic Optimal Sensor Placements**
Xianliang Li, Sheng Xu, K. C. Ho.
IEEE Transactions on Aerospace and Electronic Systems.
- 2025 **AOA Sensor Placement for Anchor-Assisted Target Localization in GNSS-Denied Environment: Formulation, Bounds and Optimization**
Sheng Xu, Linlong Wu, Xianliang Li, Xinyu Wu, Tiantian Xu.
IEEE Transactions on Mobile Computing.
- 2025 **On the Performance Analysis of Momentum Method: A Frequency Domain Perspective**
Xianliang Li, Jun Luo, Zhiwei Zheng, Hanxiao Wang, Li Luo, Lingkun Wen, Linlong Wu, Sheng Xu.
International Conference on Learning Representations (ICLR).
- 2024 **Systematical Sensor Path Optimization Solutions for AOA Target Localization Accuracy Improvement with Theoretical Analysis**
Sheng Xu, Bing Zhu, Xianliang Li, Xinyu Wu, Tiantian Xu.
IEEE Transactions on Vehicular Technology.

- 2024 **3D Source Tracking Using a Position-Unknown AOA Sensor with Measurement Drift and UAV Moving Direction Optimization**
Rongrong Xu, Sheng Xu, Kaimin Cao, **Xianliang Li**, Jialiang Wang, Wujing Cao.
IEEE International Conference on Real-time Computing and Robotics.

Professional Activities

Journal Reviewer

Digital Signal Processing (DSP).

Conference Reviewer

NeurIPS (2026), ICLR (2025).